ITA0608 – Machine Learning Lab Practicals

EXP 3 : Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Aim:

To implement and demonstrate the Decision Tree based ID3 (Iterative Dichotomiser 3) Algorithm using an appropriate dataset, construct a decision tree, and classify a new sample.

Program:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, export_text
data = pd.read_csv("play_tennis.csv")
print("Training Dataset:\n")           print(data)
le = LabelEncoder()
data_encoded = data.copy()
for column in data.columns:
    data_encoded[column] = le.fit_transform(data[column])
X = data_encoded.iloc[:, :-1]
y = data_encoded.iloc[:, -1]
clf = DecisionTreeClassifier(criterion='entropy')       clf.fit(X, y)
print("\nDecision Tree:\n")
tree_rules = export_text(clf, feature_names=list(X.columns))
print(tree_rules)
new_sample = pd.DataFrame({
    "Outlook": ["Sunny"],
    "Temperature": ["Cool"],
    "Humidity": ["High"],
    "Wind": ["Strong"] })
```

```
new_encoded = pd.DataFrame()
for column in X.columns:
    encoder = LabelEncoder()
    encoder.fit(data[column])
    new_encoded[column] = encoder.transform(new_sample[column])
prediction = clf.predict(new_encoded)
result = "Yes" if prediction[0] == 1 else "No"
print("\nNew Sample:")        print(new_sample)
print("\nPrediction:")
print("Play Tennis =", result)
```

Output:

```
Training Dataset:
...
     Outlook Temperature Humidity    Wind PlayTennis
0      Sunny         Hot     High    Weak         No
1      Sunny         Hot     High  Strong         No
2   Overcast         Hot     High    Weak        Yes
3       Rain        Mild     High    Weak        Yes
4       Rain        Cool   Normal    Weak        Yes
5       Rain        Cool   Normal  Strong         No
6   Overcast        Cool   Normal  Strong        Yes
7      Sunny        Mild     High    Weak         No
8      Sunny        Cool   Normal    Weak        Yes
9       Rain        Mild   Normal    Weak        Yes
10     Sunny        Mild   Normal  Strong        Yes
11  Overcast        Mild     High  Strong        Yes
12  Overcast         Hot   Normal    Weak        Yes
13      Rain        Mild     High  Strong         No

Decision Tree:

|--- Outlook <= 0.50
|   |--- class: 1
|--- Outlook >  0.50
|   |--- Humidity <= 0.50
|   |   |--- Outlook <= 1.50
|   |   |   |--- Wind <= 0.50
|   |   |   |   |--- class: 0
|   |   |   |--- Wind >  0.50
|   |   |   |   |--- class: 1
|   |   |--- Outlook >  1.50
|   |   |   |--- class: 0
|   |--- Humidity >  0.50
|   |   |--- Wind <= 0.50
|   |   |   |--- Outlook <= 1.50
|   |   |   |   |--- class: 0
|   |   |   |--- Outlook >  1.50
|   |   |   |   |--- class: 1
|   |   |--- Wind >  0.50
|   |   |   |--- class: 1


New Sample:
  Outlook Temperature Humidity    Wind
0   Sunny        Cool     High  Strong

Prediction:
Play Tennis = No
```

Result:

The Decision Tree based ID3 Algorithm was successfully implemented using Python.